

Problem Set 6: Due on Thursday, April 9, @noon

(100 points total)

1. **Optimal Insurance with Random Cost of Effort (20 points).** Consider a worker who has preferences given by

$$E \sum_{t=0}^{\infty} \beta^t (\ln c_t - \alpha s_t n_t),$$

where c_t and n_t are consumption and effort in period t , s_t is a random variable which equals 0 or 1, both with probability $1/2$, $\alpha > 1$ is a parameter, and $\beta \in (0, 1)$. That is, the worker has a random cost of effort. He is employed at a wage that pays 1 unit of consumption good, and is endowed with one unit of time in each period.

- (a) (3 points) Write down the worker's problem in autarky, where he has no access to insurance. Characterize his optimal consumption and effort decisions.
- (b) (6 points) Consider the problem of the principal who designs an optimal insurance contract. Formulate his recursive problem (maximizing profits subject to delivering at least a certain value to the worker) under symmetric information, i.e., when both parties can observe the realized cost of effort. Characterize properties of the optimal consumption path. (You can assume without proving it, that the value function is strictly concave in the promised value.)
- (c) (8 points) Suppose now that only the worker observes the cost of effort. Formulate the recursive problem of finding an optimal contract. Show that the symmetric-information contract from part (b) is always incentive compatible. Explain this result intuitively.
- (d) (3 points) Point out which assumption of the model is responsible for the result you found in part (c).

2. **Optimal Unemployment Compensation with Unobservable Wage Offers (30 points).** Consider an unemployed person with preferences given by

$$E \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$, $c_t \geq 0$, and the utility function $u(c)$ is strictly increasing, strictly concave, and C^2 . Each period the worker draws one offer w from a *uniform* wage

distribution on the domain $[0, 1]$. After the worker has accepted a wage offer w , he receives the wage w per period forever. He is then beyond the grasp of the unemployment insurance agency. During the unemployment spell, any consumption smoothing has to be done through the unemployment insurance agency because the worker holds no assets and cannot borrow or lend.

- (a) (5 points) Characterize the worker's optimal reservation wage when he is entitled to a time-invariant unemployment compensation b of indefinite duration.
- (b) (10 points) Characterize the optimal unemployment compensation scheme under full information. That is, we assume that the insurance agency can observe and control the unemployed worker's consumption and reservation wage.
- (c) (15 points) Characterize the optimal unemployment compensation scheme under asymmetric information where the insurance agency cannot observe wage offers, though it can observe and control the unemployed worker's consumption. Discuss the optimal time profile of the unemployed worker's consumption level. How does the worker's reservation wage change over time?

3. Optimal Contracts with Unobservable Employment Status (30 points).

Consider a worker with preferences given by

$$E \sum_{t=0}^{\infty} \beta^t \ln c_t,$$

where $c_t \geq 0$ is consumption, and $\beta \in (0, 1)$. In the beginning of each period, the worker's employment status is realized: he is either employed or unemployed, with equal probability (i.e., the employment status is i.i.d. over time). All jobs pay the same hourly wage rate w , where w is publicly known.

The worker is endowed with *one unit of time* in each period. The consumption good is non-storable, and workers have no savings and cannot borrow or lend.

There is an insurance agency that designs an optimal insurance contract for the worker. The agency can borrow/save at the interest rate $1/\beta - 1$.

Suppose that the worker's *employment status is unobservable* to the insurance agency. However, while the worker can hide that he has a job, he cannot claim that he has a job when he does not have one. In addition, the worker's *labor supply is unobservable* to the agency. Moreover, while the worker can say he has no earnings (if he claims he has no job), he cannot misreport the earnings if he claims he has a job.

- (a) (5 points) Write down the problem of finding the optimal contract, where the agency maximizes the expected present discounted value of profits subject to

delivering a certain value to the worker. To do this (in a simpler form), it is useful to answer the following questions first: What is the optimal labor supply of a worker who is

- i. employed and claims to be employed,
 - ii. unemployed,
 - iii. employed but claims to be unemployed?
- (b) (10 points) Show how current-period transfers, consumption levels, and continuation values compare across states of employment/unemployment in a given period. (Feel free to assume concavity of the value function.)

Now suppose that the worker's *labor supply is observable* by the insurance agency and contractible. The *employment status is still unobservable* (but, as before, the worker cannot claim that he has a job when he does not have one).

- (c) (10 points) Write down the problem of finding the optimal contract in this case. To do this (in a simpler way), answer the following questions first: What is the labor supply prescribed by the optimal contract to a worker who is
- i. employed and claims to be employed,
 - ii. unemployed,
 - iii. employed but claims to be unemployed?
- (d) (5 points) Show that in this setup it must be the case that the incentive constraint does not bind. Provide the intuition for this result. (*Hint: One way to show this is to take FOCs and derive that the Lagrange multiplier on the IC must be zero. Alternatively, you can simply verify that the full-insurance contract is incentive compatible.*)

4. **Optimal Contract with Unobservable Effort (20 points).** Consider an environment with a risk-neutral principal and a risk-averse agent. In each period t over an infinite horizon, the agent exerts effort to produce output, which equals $\bar{y}_h$ with probability a_t and $\bar{y}_\ell$ with probability $1 - a_t$, where $\bar{y}_h > \bar{y}_\ell > 0$ and $a_t \in [0, 1]$ is the agent's effort in that period. Let $c_t \geq 0$ denote the agent's consumption in period t .

The agent has preferences given by

$$E \sum_{t=0}^{\infty} \beta^t [u(c_t) - \psi(a_t)],$$

where $\beta \in (0, 1)$, and the functions u and ψ satisfy $u'(c) > 0$, $u''(c) < 0$ for all c , and $\psi'(a) > 0$, $\psi''(a) > 0$, $\psi'''(a) > 0$ for all a .

The principal is risk neutral and has preferences given by

$$E \sum_{t=0}^{\infty} \beta^t [y_t - c_t].$$

In what follows, assume that both parties can commit to contracts.

- (a) (13 points) Formulate the recursive problem of finding the optimal contract between the principal and the agent assuming that the agent's effort is unobservable by the principal. Derive the first-order and Envelope conditions.

How do the agent's consumption and promised values compare over time and across states? How do they behave following a sequence of high (low) output realizations? (Feel free to assume that the value function is decreasing and strictly concave, and that the solution for effort is interior.)

- (b) (7 points) Suppose that $u(c) = c$ and $\psi(a) = \gamma a^2/2$, where $\gamma > \bar{y}_h - \bar{y}_\ell$. Find the optimal contract in closed form (as a function of v). How does the implemented effort level in a given period compare to the effort level that the principal would have chosen herself if she could produce without the agent (and incurred the same cost of effort)?